

Performance Experiment

The project tests file upload/download performance and comparing the same

uploaded /downloaded file using Microsoft Graph.

There is possibility to test the files in both ways :

- Real Files (files that live on the user device)
- Generated Files(files that are generated by the application by two inputs size and type of the file)

Each file goes through the same following flow : create or select file -> hash calculation of uploaded file -> upload -> download -> hash calculation of downloaded file -> hash and file comparison -> upload/download time and speed calculation

Table 1. Generated File Transfer Performance and Integrity Validation Results

File Type	Size	Upload Time	Download Time	Upload Speed	Download Speed	Hash Check	File Compare
Txt	10240 KB	85782 ms	82247 ms	0.12 MB/s	0.12 MB/s	MATCH	MATCH
Jpg	10240 KB	99924 ms	66232 ms	0.10 MB/s	0.15 MB/s	MATCH	MATCH
Zip	10240 KB	96130 ms	209463 ms	0.10 MB/s	0.05 MB/s	MATCH	MATCH
Png	10240 KB	82730 ms	72499 ms	0.12 MB/s	0.14 MB/s	MATCH	MATCH

Table 2. Generated Text File Experiment File Results

File Type	Size	Upload Time	Download Time	Upload Speed	Download Speed	Hash Check	File Compare
Txt	10240 KB	85782 ms	82247 ms	0.12 MB/s	0.12 MB/s	MATCH	MATCH
Txt	10240 KB	258853 ms	258853 ms	0.04 MB/s	0.11 MB/s	MATCH	MATCH
Txt	10240 KB	86714 ms	82612 ms	0.12 MB/s	0.12 MB/s	MATCH	MATCH
Txt	100 KB	74960 ms	75835 ms	0.00 MB/s	0.00 MB/s	MATCH	MATCH
Txt	5120 KB	82952 ms	114117 ms	0.06 MB/s	0.04 MB/s	MATCH	MATCH

Table 3 . Real PDF File Experiment Results							
.pdf	250.11 KB	64358 ms	68919 ms	0.00 MB/s	0.00 MB/s	MATCH	MATCH
.pdf	250.11 KB	70696 ms	69524 ms	0.00 MB/s	0.00 MB/s	MATCH	MATCH
.pdf	250.11 KB	70501 ms	64924 ms	0.00 MB/s	0.00 MB/s	MATCH	MATCH
Table 4. Real Text File Experiment Results 11419KB							
Txt	11419 KB	87613 ms	70755 ms	0.13 MB/s	0.16 MB/s	MATCH	MATCH
Txt	11419 KB	79960 ms	54528 ms	0.14 MB/s	0.20 MB/s	MATCH	MATCH
Txt	11419 KB	84952 ms	57980 ms	0.13 MB/s	0.19 MB/s	MATCH	MATCH
Table 5. Generated Text File Experiment Results 11419KB							
Txt	11419 KB	69363 ms	60568 ms	0.16 MB/s	0.18 MB/s	MATCH	MATCH
Txt	11419 KB	80398 ms	71873 ms	0.14 MB/s	0.16 MB/s	MATCH	MATCH
Txt	11419 KB	70696 ms	69524 ms	0.12 MB/s	0.18 MB/s	MATCH	MATCH

Performance Behaviour

Smaller and larger files often have similar transfer durations. File transfer is influenced more by system and network overhead than by file size alone

Generated vs Real Files

Generated files have more stable and predictable results because they are created in a controlled environment with structure and size.

Real files show more variation in performance, due to differences in file structure and how cloud services process different file formats.

File Type Impact

Different file types show different performance behavior even though all files are transmitted as binary data.

This suggests that cloud processing can affect transfer speed.